

Problem Set 4:

Abgabe: 12:00, Freitag, 08. Mai 2026; Lösungen: Freitag, 22. Mai 2026

Problem 1 Point charges in an infinite dielectric

Let the half-space $z < 0$ be filled with a homogeneous and isotropic dielectric with dielectric constant ϵ/ϵ_0 , whereas the half-space $z > 0$ is a vacuum. A point charge q in the vacuum is placed at the point $\mathbf{r}_0 = (0, 0, D)^T$, and we wish to determine the electrostatic potential throughout the entire space.

Using the method of mirror charges, show that for $z > 0$ the solution is of the form

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{|\mathbf{r} - \mathbf{r}_0|} + \frac{1}{4\pi\epsilon_0} \frac{q'}{|\mathbf{r} + \mathbf{r}_0|} \quad (1)$$

whereas for $z < 0$ an approximation of the form

$$\tilde{\phi}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q''}{|\mathbf{r} - \mathbf{r}_0|} \quad (2)$$

suffices. Determine the mirror charges q' and q'' .

Solution:

If the ansatz is flexible enough to fulfill the boundary conditions at the boundary, it is justified a posteriori by the existence of a solution for q' and q'' . Choose cylindrical coordinates, where

$$x^2 + y^2 = \rho^2. \quad (3)$$

Furthermore, define

$$R_{\pm} := \sqrt{\rho^2 + (z \mp D)^2}. \quad (4)$$

Then

$$\phi_{>}(\rho, z) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{R_{+}} + \frac{q'}{R_{-}} \right), \quad (5)$$

and

$$\phi_{<}(\rho, z) = \frac{1}{4\pi\epsilon_0} \frac{q''}{R_{+}}. \quad (6)$$

The boundary conditions at the boundary are the continuity of $\mathbf{E}_{\parallel}$ and $\hat{\mathbf{n}} \cdot \mathbf{D}$. It follows that

$$\epsilon_0 E_{>}^z = \epsilon \epsilon_0 E_{<}^z, \quad (7)$$

and

$$E_{>}^{\rho} = E_{<}^{\rho}, \quad (8)$$

must hold at $z = 0$. Using

$$\frac{\partial R_{\pm}}{\partial z} = \frac{z \mp D}{R_{\pm}}, \quad (9)$$

we find

$$\left(-q \frac{z-D}{R_+^3} - q' \frac{z+D}{R_-^3} \right) \Big|_{z=0} = -\epsilon q'' \frac{z-D}{R_+^3} \Big|_{z=0}. \quad (10)$$

Therefore,

$$q - q' = \epsilon q''. \quad (11)$$

On the other hand,

$$E_{>}^\rho = E_{<}^\rho \iff \frac{\partial \phi_{>}}{\partial \rho} \Big|_{z=0} = \frac{\partial \phi_{<}}{\partial \rho} \Big|_{z=0}. \quad (12)$$

This immediately implies

$$q + q' = q''. \quad (13)$$

Rearranging,

$$q - q' = \epsilon(q + q'), \quad (14)$$

$$\implies q' = \frac{1-\epsilon}{1+\epsilon} q. \quad (15)$$

It follows that

$$q'' = \left(\frac{1-\epsilon}{1+\epsilon} + 1 \right) q = \frac{2}{1+\epsilon} q. \quad (16)$$

We have thus found a valid solution for the image charges that satisfies the boundary conditions, which justifies our ansatz. Notice that our solution recovers the field of a single point charge in vacuum in the limit $\epsilon \rightarrow 1$.

Problem 2 (Zentralübung) Minimizing the Energy Functional

Consider a volume V on the surface of which boundary conditions determine the value of $\hat{\mathbf{n}} \cdot \mathbf{D}$.

(2.a) Use the Lagrange multiplier method to show that the functional

$$U_E[\mathbf{D}] = \frac{1}{2} \int_V d^3\mathbf{r} |\mathbf{D}(\mathbf{r})|^2 / \epsilon \quad (17)$$

(where all vector fields $\mathbf{D}(\mathbf{r})$ satisfy the relation $\text{div } \mathbf{D} = \rho_f$) takes on an extremum for exactly those functions $\mathbf{D}(\mathbf{r})$ where

$$\mathbf{D}(\mathbf{r}) = \epsilon \mathbf{E}(\mathbf{r}). \quad (18)$$

(2.b) Now show that the extremum found is a minimum.

Hint: Since $\text{div } \mathbf{D}(\mathbf{r}) = \rho_f(\mathbf{r})$ holds pointwise, it is useful to introduce a Lagrangian function $\varphi(\mathbf{r})$ to enforce the boundary condition throughout V .

Solution:

(2.a) We introduce a scalar Lagrange multiplier field $\lambda(\mathbf{r})$ and consider the augmented functional

$$F[\mathbf{D}, \lambda] = \int_V d^3\mathbf{r} \left(\frac{1}{2\epsilon} |\mathbf{D}(\mathbf{r})|^2 - \lambda(\mathbf{r}) (\nabla \cdot \mathbf{D}(\mathbf{r}) - \rho_f) \right). \quad (19)$$

To extremize the functional above, we want to take the variation of F with respect to its arguments, i.e. the fields $\mathbf{D}(\mathbf{r})$ and $\lambda(\mathbf{r})$ independently. Due to the boundary conditions, we have to restrict the variation $\delta\mathbf{D}(\mathbf{r})$ s.t.

$$\hat{\mathbf{n}} \cdot \delta\mathbf{D}(\mathbf{r}) = 0 \quad \forall \mathbf{r} \in \partial V. \quad (20)$$

We obtain

$$\delta F[\mathbf{D}, \lambda] = \int_V d^3\mathbf{r} \left(\frac{1}{\epsilon} \mathbf{D}(\mathbf{r}) \delta\mathbf{D}(\mathbf{r}) - \lambda(\mathbf{r}) \nabla \cdot \delta\mathbf{D}(\mathbf{r}) - \delta\lambda(\mathbf{r}) (\nabla \cdot \mathbf{D}(\mathbf{r}) - \rho_f) \right). \quad (21)$$

By Gauss Theorem, the second part is equivalent to

$$- \int_V d^3\mathbf{r} \lambda(\mathbf{r}) \nabla \cdot \delta\mathbf{D}(\mathbf{r}) = \int_V d^3\mathbf{r} \nabla \lambda(\mathbf{r}) \delta\mathbf{D}(\mathbf{r}) - \underbrace{\int_{\partial V} d^2S \hat{\mathbf{n}} \cdot \delta\mathbf{D}(\mathbf{r})}_{=0}. \quad (22)$$

Since we can vary the fields $\mathbf{D}$ and λ independently, we get

$$\frac{1}{\epsilon} \mathbf{D}(\mathbf{r}) + \nabla \lambda(\mathbf{r}) = 0 \quad (23)$$

$$\nabla \cdot \mathbf{D}(\mathbf{r}) - \rho_f = 0. \quad (24)$$

Plugging the first into the second equations yields

$$-\epsilon \Delta \lambda = \rho_f = -\epsilon \Delta \phi, \quad (25)$$

where ϕ is the electric potential. With this we find

$$\lambda = \phi + f, \quad (26)$$

where f is an arbitrary function with $\Delta f = 0$ and $\hat{\mathbf{n}} \cdot \nabla f = 0$ on ∂V .

$$\mathbf{D}(\mathbf{r}) = -\epsilon \nabla \lambda = -\epsilon \nabla \phi - \epsilon \nabla f = \epsilon \mathbf{E} - \epsilon \nabla f. \quad (27)$$

So indeed $\mathbf{D}(\mathbf{r}) = \epsilon \mathbf{E}$ is an extrema of the functional U_E fulfilling the constraint $\nabla \cdot \mathbf{D}(\mathbf{r}) = \rho_f$.

(2.b) To show that $\mathbf{D}(\mathbf{r}) = \epsilon \mathbf{E}$ is actually a minimum, we assume that $\mathbf{D}'$ is another field fulfilling the same constraints

$$\nabla \cdot \mathbf{D}'(\mathbf{r}) = \rho_f, \quad \hat{\mathbf{n}} \cdot \mathbf{D}'(\mathbf{r}) = \hat{\mathbf{n}} \cdot \mathbf{D}(\mathbf{r}) \quad \forall \mathbf{r} \in \partial V. \quad (28)$$

Defining $\mathbf{u} := \mathbf{D}' - \mathbf{D}$, we find that

$$U_E[\mathbf{D}'] = U_E[\mathbf{D} + \mathbf{u}] \quad (29)$$

$$= \int_V d^3\mathbf{r} \frac{1}{2\epsilon} |\mathbf{D}(\mathbf{r}) + \mathbf{u}(\mathbf{r})|^2 \quad (30)$$

$$= \int_V d^3\mathbf{r} \frac{1}{2\epsilon} (|\mathbf{D}(\mathbf{r})|^2 + |\mathbf{u}(\mathbf{r})|^2 + 2\mathbf{D}(\mathbf{r})\mathbf{u}(\mathbf{r})) \quad (31)$$

$$= \int_V d^3\mathbf{r} \frac{1}{2\epsilon} (|\mathbf{D}(\mathbf{r})|^2 + |\mathbf{u}(\mathbf{r})|^2) \quad (32)$$

$$\geq U_E[\mathbf{D}]. \quad (33)$$

Here, we have used that the mixed term

$$2 \int_V d^3\mathbf{r} \mathbf{D}(\mathbf{r}) \mathbf{u}(\mathbf{r}) = -2\epsilon \int_V d^3\mathbf{r} \nabla \phi(\mathbf{r}) \mathbf{u}(\mathbf{r}) \quad (34)$$

$$= 2\epsilon \left[\int_V d^3\mathbf{r} \phi(\mathbf{r}) \nabla \cdot \mathbf{u}(\mathbf{r}) - \int_{\partial V} d^2\mathbf{S} \nabla \phi(\mathbf{r}) \mathbf{u}(\mathbf{r}) \right] \quad (35)$$

$$= 0 \quad (36)$$

since

$$\nabla \cdot \mathbf{u}(\mathbf{r}) = 0 \quad \text{and} \quad \mathbf{u}(\mathbf{r}) \Big|_{\mathbf{r} \in \partial V} = 0. \quad (37)$$

Thus $\mathbf{D}(\mathbf{r}) = \epsilon \mathbf{E}$ is indeed a minimum of the functional U_E .

Problem 3 A Hole in a Radially Polarized Medium

The polarization throughout the entire space has the form

$$\mathbf{P}(\mathbf{r}) = P \Theta(r - R) \hat{\mathbf{r}}. \quad (38)$$

P and R are constants, and Θ denotes the Heaviside distribution. Determine the polarization charge density and the electric field in the entire space.

Solution:

Let

$$\mathbf{P}(\mathbf{r}') = P \Theta(r' - R) \hat{\mathbf{r}}'. \quad (39)$$

The polarization charge density is

$$\rho_{\text{pol}} = -\nabla' \cdot \mathbf{P}. \quad (40)$$

In spherical coordinates, the divergence of a radial vector field is given by

$$\nabla' \cdot (f(r') \hat{\mathbf{r}}') = \frac{1}{r'^2} \frac{\partial}{\partial r'} (r'^2 f(r')). \quad (41)$$

Thus,

$$\begin{aligned} \rho_{\text{pol}} &= -\nabla' \cdot \mathbf{P} \\ &= -\frac{P}{r'^2} \frac{\partial}{\partial r'} (r'^2 \Theta(r' - R)) \\ &= -\frac{P}{r'^2} (2r' \Theta(r' - R) + r'^2 \delta(r' - R)) \\ &= -\frac{2P}{r'} \Theta(r' - R) - P \delta(r' - R). \end{aligned} \quad (42)$$

Now we use Gauß's law:

$$\nabla' \cdot \mathbf{E}(\mathbf{r}') = \frac{\rho_{\text{f}}(\mathbf{r}') + \rho_{\text{pol}}(\mathbf{r}')}{\epsilon_0}, \quad (43)$$

where $\rho_{\text{f}}(\mathbf{r}') = 0$ in our problem, since there is no free charge. Equivalently,

$$\oint_{\partial V} d\mathbf{A}' \cdot \mathbf{E} = \frac{1}{\epsilon_0} \int_V dV' \rho_{\text{pol}}. \quad (44)$$

By spherical symmetry, we have

$$\mathbf{E}(\mathbf{r}') = E(r') \hat{\mathbf{r}}'. \quad (45)$$

For $r < R$, no polarization charge is enclosed, hence

$$\mathbf{E}(\mathbf{r}) = \mathbf{0}, \quad r < R. \quad (46)$$

For $r > R$, the enclosed polarization charge is, defining

$$B_r := \{\mathbf{r}' \in \mathbb{R}^3 \mid |\mathbf{r}'| < r\}, \quad (47)$$

given by

$$\begin{aligned} Q_r &= \int_{B_r} d^3\mathbf{r}' \rho_{\text{pol}}(\mathbf{r}') \\ &= \int_{B_r} d^3\mathbf{r}' \left[-\frac{2P}{r'} \Theta(r' - R) - P\delta(r' - R) \right] \\ &= -8\pi P \int_R^r dr' r' - 4\pi R^2 P \\ &= -4\pi P(r^2 - R^2) - 4\pi R^2 P \\ &= -4\pi P r^2. \end{aligned} \quad (48)$$

Gauß's law gives

$$E(r) 4\pi r^2 = -\frac{4\pi P r^2}{\varepsilon_0}, \quad (49)$$

and therefore

$$E(r) = -\frac{P}{\varepsilon_0}. \quad (50)$$

In summary,

$$\mathbf{E}(\mathbf{r}) = -\frac{P}{\varepsilon_0} \Theta(r - R) \hat{\mathbf{r}}. \quad (51)$$

Problem 4 Fourier Decomposition of the Green's Function in Vacuum

For $\mathbf{r} \in \mathbb{R}^3$, we decompose the δ -distribution into a longitudinal and a transverse component:

$$\delta(\mathbf{r} - \mathbf{r}') = \delta(\mathbf{r}_\perp - \mathbf{r}'_\perp) \delta(z - z').$$

Using this, show by direct integration the following relation for the Green's function of the electrostatic potential in a vacuum:

$$G_0(\mathbf{r}, \mathbf{r}') = \frac{1}{2\varepsilon_0} \int \frac{d^2\mathbf{k}_\perp}{(2\pi)^2} e^{i\mathbf{k}_\perp \cdot (\mathbf{r}_\perp - \mathbf{r}'_\perp)} \frac{1}{k_\perp} e^{-k_\perp |z - z'|}.$$

Solution:

We decompose

$$\delta^{(3)}(\mathbf{r} - \mathbf{r}') = \delta^{(2)}(\mathbf{r}_\perp - \mathbf{r}'_\perp) \delta(z - z'). \quad (52)$$

The Green's function satisfies

$$\nabla^2 G_0(\mathbf{r}, \mathbf{r}') = -\frac{1}{\varepsilon_0} \delta^{(3)}(\mathbf{r} - \mathbf{r}'). \quad (53)$$

We Fourier transform in the plane spanned by $\{\mathbf{r}_\perp, \mathbf{r}'_\perp\}$. Hence

$$G_0(\mathbf{r}, \mathbf{r}') = \int \frac{d^2 \mathbf{k}_\perp}{(2\pi)^2} e^{i\mathbf{k}_\perp \cdot (\mathbf{r}_\perp - \mathbf{r}'_\perp)} g(\mathbf{k}_\perp; z, z'). \quad (54)$$

Then

$$\nabla^2 \longrightarrow \frac{\partial^2}{\partial z^2} - k_\perp^2, \quad (55)$$

as seen before in problem set 01, exercise 02. Thus g satisfies

$$\left(\frac{\partial^2}{\partial z^2} - k_\perp^2 \right) g(\mathbf{k}_\perp; z, z') = -\frac{1}{\varepsilon_0} \delta(z - z'). \quad (56)$$

Equivalently, after translating the source to the origin,

$$(\partial_z^2 - k_\perp^2) g(\mathbf{k}_\perp; z - z', 0) = -\frac{1}{\varepsilon_0} \delta(z - z'). \quad (57)$$

Fourier transforming, we immediately find that

$$g(\mathbf{k}_\perp; z - z', 0) = \frac{1}{\varepsilon_0} \mathcal{F}_{k_z \rightarrow z - z'}^{-1} \left[\frac{1}{k_\perp^2 + k_z^2} \right] = \frac{1}{2\varepsilon_0 k_\perp} e^{-k_\perp |z - z'|}. \quad (58)$$

Here we used the Fourier transform of a Lorentzian function. It follows that

$$G_0(\mathbf{r}, \mathbf{r}') = \frac{1}{2\varepsilon_0} \int \frac{d^2 \mathbf{k}_\perp}{(2\pi)^2} e^{i\mathbf{k}_\perp \cdot (\mathbf{r}_\perp - \mathbf{r}'_\perp)} \frac{1}{k_\perp} e^{-k_\perp |z - z'|}, \quad (59)$$

as desired.